

Block-Adaptive Autocorrelation Estimation for One-Bit Quantized Gaussian Data

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Abstract—Adaptive estimation of the autocorrelation function (ACF) plays a crucial role in statistical signal processing for non-stationary environments. Many conventional ACF estimators are developed for unquantized signals. In practice, these signals are replaced with high-resolution quantized signals to mitigate model mismatch. However, this approach increases hardware costs and power consumption due to high-resolution analog-to-digital converters (ADCs). This situation can be avoided by using one-bit ADCs. This paper proposes an adaptive ACF estimator for one-bit quantized data. The proposed estimator is founded on block-adaptive estimators for the mean and autocorrelation matrix of the one-bit data. The mismatches from one-bit quantization are alleviated via optimization problems for the power and normalized ACF estimation of the unquantized data. These optimization problems are reformulated in terms of auxiliary variables to remove the constraints on the power and normalized ACF. This approach also enables gradient-descent methods with closed-form expressions. Numerical examples demonstrate that the proposed method with one-bit data follows a similar trend to the conventional ACF estimation with unquantized data in the stationary and non-stationary environments.

Index Terms—One-bit quantization, one-bit Hermite law, gradient descent, adaptive parameter estimation.

I. INTRODUCTION

Estimating the autocorrelation function (ACF) of a random process has been a central topic in statistical signal processing [2]–[4]. The ACF embeds parameters in random processes, such as autoregressive moving-average (ARMA) parameters in ARMA processes [2]–[4] and frequency components in harmonic processes [2]–[4]. Techniques for ACF estimation can be extended to covariance estimation, power estimation, and direction-of-arrival (DOA) estimation [2]–[6]. However, many ACF estimators are developed for stationary random processes.

Practical scenarios are *non-stationary*, where the signal characteristics change over time. Furthermore, only a short frame of recent data is available to parameter estimators. In this setting, *adaptive methods* update the signal parameters based on the past estimates and the current observation [7], [8]. These methods can feature *low computational cost*, *fast tracking response*, and *low estimation errors*.

Conventional adaptive methods are primarily developed for original (unquantized) data [7], [8]. In digital signal processing

[9], these data are quantized to high-resolution samples. This approach requires high-resolution analog-to-digital converters (ADCs), which increase cost and power consumption. For this reason, *one-bit ADCs* [10] with a quantization threshold λ map each real-valued sample to binary values, which find applications in hardware-constrained scenarios [11]–[14]. However, the one-bit samples deviate from the original (unquantized) signals, resulting in significant quantization errors and degraded performance of conventional adaptive methods.

For ACF estimation using one-bit data, several related works [15]–[21] assume that the original Gaussian data is processed by one-bit quantizers with $\lambda \neq 0$. In this setting, the power can be estimated through the inverse Q -function [15]–[20]. The second-order statistics of the original data can be recovered via the modified arcsine law [16]–[18], the one-bit Hermite law (OBHL) [19], [20], and maximum-likelihood methods [21]. These methods may involve computationally-intensive steps. The approximated OBHL (AppOBHL) [20] offers closed-form expressions, low computational cost, and provable bounds on approximation errors. These features make it suitable for adaptive one-bit ACF estimation in non-stationary environments.

This paper proposes an adaptive ACF estimator for one-bit quantized Gaussian data. These data are divided into blocks, from which the one-bit mean and autocorrelation matrix (ACM) are estimated adaptively. These quantities enable optimization problems for power and ACF estimation. The positivity of the power estimates and the boundedness of the normalized ACF are transformed into unconstrained auxiliary variables. These auxiliary variables enable a gradient-descent-based approach with closed-form update equations. Note that this paper primarily focuses on one-bit quantizers. Extensions to low-resolution quantizers [22], [23] are left for future work.

Paper Outline: Section II reviews the AppOBHL. Section III presents the adaptive mean and ACM estimators. Section IV proposes the ACF estimator for one-bit Gaussian data. Section V demonstrates numerical examples for the estimation performance, while Section VI concludes this paper.

II. PRELIMINARIES

Let $x(n)$ be a real-valued scalar Gaussian process with zero mean. The one-bit quantized signal $y(n)$ is defined as

$$y(n) \triangleq \mathcal{Q}_1(x(n), \lambda) = \begin{cases} 1, & \text{if } x(n) \geq \lambda, \\ -1, & \text{if } x(n) < \lambda, \end{cases} \quad (1)$$

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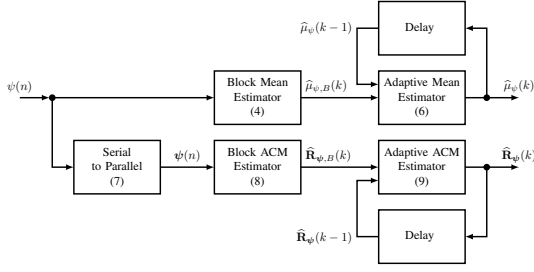


Fig. 1. A block diagram of the adaptive mean and ACM estimators.

where $\mathcal{Q}_1(\cdot, \lambda)$ denotes a one-bit quantizer with a quantization threshold λ . For time indices n_1 and n_2 , the vector $[x(n_1), x(n_2)]^T$ follows a bivariate normal distribution with zero mean. This distribution is parametrized by the parameter vector $\theta \triangleq [\sigma_1, \sigma_2, \rho_x]^T$, where $\sigma_1 \triangleq \sqrt{\mathbb{E}[x(n_1)^2]}$ and $\sigma_2 \triangleq \sqrt{\mathbb{E}[x(n_2)^2]}$ are the standard deviations (SDs) of $x(n_1)$ and $x(n_2)$, respectively. The correlation coefficient between $x(n_1)$ and $x(n_2)$ is $\rho_x \triangleq \mathbb{E}[x(n_1)x(n_2)]/(\sigma_1\sigma_2)$.

According to (1), the mean of $y(n_1)$ becomes [15], [18].

$$\mathbb{E}[y(n_1)] = \mathcal{M}(\sigma_1, \lambda) \triangleq -1 + 2Q(\gamma_1). \quad (2)$$

The Q function [24] is denoted by $Q(\cdot)$. The normalized quantization threshold is $\gamma_1 \triangleq \lambda/\sigma_1$. For a similar reason, we have $\mathbb{E}[y(n_2)] = \mathcal{M}(\sigma_2, \lambda) = -1 + 2Q(\gamma_2)$ with $\gamma_2 \triangleq \lambda/\sigma_2$.

The one-bit correlation $\mathbb{E}[y(n_1)y(n_2)]$ can be expressed in the OBHL [19]. However, the expression of the OBHL contains a summation with infinite terms. For computational purposes, the AppOBHL introduces the amplitude parameter A , the decay parameter β , and the truncation parameter L , and defines a vector $\phi \triangleq [A, \beta, L]^T$. In [20], the one-bit correlation $\mathbb{E}[y(n_1)y(n_2)]$ is approximated by the expression $\mathcal{G}_{\text{AppOBHL}}(\theta, \lambda, \phi)$

$$\mathcal{G}_{\text{AppOBHL}}(\theta, \lambda, \phi) = \mathcal{M}(\sigma_1, \lambda)\mathcal{M}(\sigma_2, \lambda) + A\rho_x\mathcal{S}(\theta, \beta\rho_x) + \frac{2\rho_x \exp(-\bar{\gamma}^2)}{\pi} \sum_{n=0}^{L-1} e(n)d(n)\rho_x^n. \quad (3)$$

The expression $\mathcal{S}(\theta, \eta) \triangleq \frac{2}{\pi\sqrt{1-\eta^2}} \exp\left(-\frac{\bar{\gamma}^2 - \eta\gamma_1\gamma_2}{1-\eta^2}\right)$, where $\bar{\gamma} \triangleq \sqrt{(\gamma_1^2 + \gamma_2^2)/2}$. The sequences $e(n)$ and $d(n)$ are defined as $e(n) \triangleq \frac{1}{n+1} - A\beta^n$ and $d(n) \triangleq \frac{1}{2^n n!} H_n\left(\frac{\gamma_1}{\sqrt{2}}\right) H_n\left(\frac{\gamma_2}{\sqrt{2}}\right)$. The Hermite polynomial of order n is defined as $H_n(u) \triangleq (-1)^n \exp(u^2) \left(\frac{d}{du}\right)^n \exp(-u^2)$.

III. BLOCK-ADAPTIVE MEAN AND ACM ESTIMATORS

Fig. 1 depicts a block diagram of the adaptive mean and ACM estimators. The input signal $\psi(n)$ is defined for $n \in [0, N-1] \triangleq \{0, 1, \dots, N-1\}$. The time index n is decomposed into $n = kB + \ell$, where B is the block size, k is the block index, and $\ell \in [0, B-1]$. There are $K \triangleq N/B$ blocks. For the generality of presentation, the input signal $\psi(n)$ can be the original data $x(n)$ or the one-bit quantized data $y(n)$.

The block mean of $\psi(n)$ in the k th block is estimated by

$$\hat{\mu}_{\psi,B}(k) \triangleq \frac{1}{B} \sum_{\ell=0}^{B-1} \psi(kB + \ell). \quad (4)$$

Next, the adaptive mean estimator of $\psi(n)$ is exponentially weighted by the block mean $\hat{\mu}_{\psi,B}(k)$ by [1]

$$\hat{\mu}_{\psi}(k) \triangleq \frac{1}{T_k} \sum_{k'=0}^k \tau^{k-k'} \hat{\mu}_{\psi,B}(k'), \quad (5)$$

where $T_k \triangleq \sum_{k'=0}^k \tau^{k-k'}$. The exponential parameter $\tau \in [0, 1]$. It can be shown that $T_k = \tau T_{k-1} + 1$. Due to (5), the adaptive estimator $\hat{\mu}_{\psi}(k)$ is implemented recursively by [1]:

$$\hat{\mu}_{\psi}(k) = \left(1 - \frac{1}{T_k}\right) \hat{\mu}_{\psi}(k-1) + \left(\frac{1}{T_k}\right) \hat{\mu}_{\psi,B}(k). \quad (6)$$

The ACM can be estimated in blocks. First, the input vector $\psi(n)$ for the input signal $\psi(n)$ is defined as

$$\psi(n) = [\psi(n) \quad \psi(n-1) \quad \dots \quad \psi(n-P)]^T, \quad (7)$$

where P denotes the maximum lag in estimating the ACF. Based on a similar concept to (4), the ACM in the k th block is estimated by the sample correlation matrix [1], [7]

$$\hat{\mathbf{R}}_{\psi,B}(k) \triangleq \frac{1}{B} \sum_{\ell=0}^{B-1} \psi(kB + \ell) \psi^T(kB + \ell). \quad (8)$$

The adaptive ACM estimator is defined as $\hat{\mathbf{R}}_{\psi}(k) \triangleq \frac{1}{T_k} \sum_{k'=0}^k \tau^{k-k'} \hat{\mathbf{R}}_{\psi,B}(k')$. This adaptive ACM estimator also admits a recursive implementation [1]:

$$\hat{\mathbf{R}}_{\psi}(k) = \left(1 - \frac{1}{T_k}\right) \hat{\mathbf{R}}_{\psi}(k-1) + \left(\frac{1}{T_k}\right) \hat{\mathbf{R}}_{\psi,B}(k). \quad (9)$$

The adaptive ACM estimator $\hat{\mathbf{R}}_{\psi}(k)$ can be used to estimate the ACF [1], [2]. Averaging the (off-)diagonal entries in $\hat{\mathbf{R}}_{\psi}(k)$ yields the estimated ACF of $\psi(n)$ in the k th block:

$$\hat{r}_{\psi,i}(k) \triangleq \frac{1}{P+1-i} \sum_{(p,q) \in \mathbb{M}(i)} [\hat{\mathbf{R}}_{\psi}(k)]_{p,q}, \quad (10)$$

where the lag $i \in [0, P]$. The set $\mathbb{M}(i)$ is defined as $\mathbb{M}(i) \triangleq \{(p, q) \in [1, P+1]^2 \mid p - q = i\}$, which identifies the entries in $\hat{\mathbf{R}}_{\psi}(k)$ associated with the lag i in the ACF. The (p, q) -th entry of $\hat{\mathbf{R}}_{\psi}(k)$ is denoted by $[\hat{\mathbf{R}}_{\psi}(k)]_{p,q}$.

If $\psi(n) = x(n)$, then the diagram in Fig. 1 infers $\hat{\mu}_x(k)$ and $\hat{\mathbf{R}}_x(k)$, which are the statistics of the original data. However, if $\psi(n) = y(n)$, the outputs $\hat{\mu}_y(k)$ and $\hat{\mathbf{R}}_y(k)$ deviate from $\hat{\mu}_x(k)$ and $\hat{\mathbf{R}}_x(k)$.

IV. ONE-BIT ACF ESTIMATION

The one-bit mean $\hat{\mu}_y(k)$ can be matched with the expression in (2). This concept motivates the optimization problem [19]:

$$\min_{\xi_0 > 0} (\mathcal{M}(\xi_0, \lambda) - \hat{\mu}_y(k))^2, \quad (11)$$

where the optimization variable ξ_0 has the physical meaning of the SD of $x(n)$ in the k th block.

Equation (11) can be solved using *gradient descent* [25], in which the solution is updated iteratively based on gradient. However, the *positivity* of ξ_0 in (11) needs to be handled in the gradient descent. For this reason, ξ_0 is transformed into an unconstrained auxiliary variable, ν_0 , by

$$\xi_0 = \exp(\nu_0). \quad (12)$$

Note that (12) guarantees the positivity of ξ_0 for all $\nu_0 \in \mathbb{R}$.

The variable transformation in (12) rewrites (11) into

$$\min_{\nu_0 \in \mathbb{R}} F_\mu(\nu_0, k), \quad (13)$$

where the objective function $F_\mu(\nu_0, k)$ is

$$F_\mu(\nu_0, k) \triangleq (\mathcal{M}(\exp(\nu_0), \lambda) - \hat{\mu}_y(k))^2. \quad (14)$$

Applying the gradient descent leads to the update equation:

$$\hat{\nu}_0(k) \triangleq \hat{\nu}_0(k-1) - \alpha_0 \left(\frac{\partial F_\mu(\nu_0, k)}{\partial \nu_0} \right) \Big|_{\nu_0 = \hat{\nu}_0(k-1)}, \quad (15)$$

where $\hat{\nu}_0(k)$ denotes the estimate of ν_0 in the k th block. The step-size parameter is $\alpha_0 > 0$. According to (14), the derivative $\frac{\partial F_\mu(\nu_0, k)}{\partial \nu_0}$ can be expanded into

$$\frac{\partial F_\mu(\nu_0, k)}{\partial \nu_0} = 2(\mathcal{M}(\xi_0, \lambda) - \hat{\mu}_y(k)) \frac{\partial \mathcal{M}(\xi_0, \lambda)}{\partial \xi_0} \frac{\partial \xi_0}{\partial \nu_0}. \quad (16)$$

Based on (2), the derivative $\frac{\partial \mathcal{M}(\xi_0, \lambda)}{\partial \xi_0}$ becomes

$$\frac{\partial \mathcal{M}(\xi_0, \lambda)}{\partial \xi_0} = \sqrt{\frac{2}{\pi}} \left(\frac{\lambda}{\xi_0^2} \right) \exp \left(-\frac{1}{2} \left(\frac{\lambda}{\xi_0} \right)^2 \right). \quad (17)$$

From (12), the derivative $\frac{\partial \xi_0}{\partial \nu_0} = \xi_0$. Based on (12), the SD and power of $x(n)$ in the k th block can be estimated by

$$\hat{\xi}_0(k) \triangleq \exp(\hat{\nu}_0(k)), \quad \hat{r}_{x,0}(k) \triangleq (\hat{\xi}_0(k))^2. \quad (18)$$

The ACF $r_{x,i}$ is next estimated from the adaptive ACM estimator $\hat{\mathbf{R}}_y(k)$ in Fig. 1 with $\psi(n) = y(n)$ and $\psi(n) = \mathbf{y}(n)$. It can be shown [7] that the ACM $\mathbb{E}[\mathbf{y}(n)\mathbf{y}^T(n)]$ is a *Hermitian Toeplitz matrix* for stationary $y(n)$. Furthermore, the diagonal entries of $\mathbb{E}[\mathbf{y}(n)\mathbf{y}^T(n)]$ are 1 since $y(n) \in \{1, -1\}$. These structures yield the following optimization problem:

$$\min_{\boldsymbol{\xi} \in (-1, 1)^P} \left\| \left(\mathbf{I} + \sum_{i=1}^P c_{y,i} \mathbf{E}_i \right) - \hat{\mathbf{R}}_y(k) \right\|_F^2, \quad (19)$$

where the vector $\boldsymbol{\xi}$ is

$$\boldsymbol{\xi} \triangleq [\xi_1 \quad \xi_2 \quad \dots \quad \xi_P]^T. \quad (20)$$

The optimization variable ξ_i shares the physical meaning of the normalized autocorrelation function of $x(n)$, defined as $\mathbb{E}[x(n)x(n-i)]/\mathbb{E}[x^2(n)]$. The notation $c_{y,i}$ has the interpretation of the one-bit correlation $\mathbb{E}[y(n)y(n-i)]$. For computational purposes, $c_{y,i}$ is set to the AppOBHL in (3):

$$c_{y,i} = \mathcal{G}_{\text{AppOBHL}}(\boldsymbol{\theta}_i(k), \lambda, \phi), \quad (21)$$

where parameter vector $\boldsymbol{\theta}_i(k) \triangleq [\hat{\xi}_0(k), \hat{\xi}_0(k), \xi_i]^T$. The matrix $\mathbf{E}_i$ specifies the Hermitian Toeplitz structure of the

ACM. For $(p, q) \in \llbracket 1, P+1 \rrbracket^2$, the (p, q) -entry of $\mathbf{E}_i$ is 1 if $|p - q| = i$ and 0 otherwise. The Frobenius norm of a matrix $\mathbf{M}$ is represented by $\|\mathbf{M}\|_F$.

The optimization variable ξ_i in (19) lies in the bounded interval $(-1, 1)$. This interval can be mapped into the real axis $\mathbb{R}$ for the applicability of gradient descent. Let $\nu_i \in \mathbb{R}$ be an auxiliary variable for $i \in \llbracket 1, P \rrbracket$. The optimization variable ξ_i is associated with ν_i by

$$\xi_i = \frac{1 - \exp(-\nu_i)}{1 + \exp(-\nu_i)}. \quad (22)$$

Based on the variable transformation in (22), the optimization problem in (19) is modified into

$$\min_{\boldsymbol{\nu} \in \mathbb{R}^P} F_{\mathbf{R}}(\boldsymbol{\nu}, k), \quad (23)$$

where $\boldsymbol{\nu} \triangleq [\nu_1, \nu_2, \dots, \nu_P]^T$. The objective function $F_{\mathbf{R}}(\boldsymbol{\nu}, k)$ is defined as

$$F_{\mathbf{R}}(\boldsymbol{\nu}, k) = \left\| \left(\mathbf{I} + \sum_{i=1}^P c_{y,i} \mathbf{E}_i \right) - \hat{\mathbf{R}}_y(k) \right\|_F^2. \quad (24)$$

For simplicity, we define a vector for the one-bit correlation

$$\mathbf{c} \triangleq [c_{y,1} \quad c_{y,2} \quad \dots \quad c_{y,P}]^T. \quad (25)$$

Denoting $\text{vec}(\cdot)$ as the vectorization operator and $\|\cdot\|_2$ as the ℓ_2 norm, we have $\|\mathbf{M}\|_F = \|\text{vec}(\mathbf{M})\|_2$ for a matrix $\mathbf{M}$. This property rewrites the objective function $F_{\mathbf{R}}(\boldsymbol{\nu}, k)$ as

$$F_{\mathbf{R}}(\boldsymbol{\nu}, k) = \|\mathbf{E}\mathbf{c} - \mathbf{b}\|_2^2, \quad (26)$$

where the matrix $\mathbf{E}$ and vector $\mathbf{b}$ are defined as

$$\mathbf{E} \triangleq [\text{vec}(\mathbf{E}_1) \quad \text{vec}(\mathbf{E}_2) \quad \dots \quad \text{vec}(\mathbf{E}_P)], \quad (27)$$

$$\mathbf{b} \triangleq \text{vec}(\hat{\mathbf{R}}_y(k) - \mathbf{I}). \quad (28)$$

The optimization problem (23) reduces to an *unconstrained* one. This formulation is compatible with gradient descent. We update the auxiliary parameter vector $\boldsymbol{\nu}$ in the k th block by

$$\hat{\boldsymbol{\nu}}(k) = \hat{\boldsymbol{\nu}}(k-1) - \alpha_{\boldsymbol{\nu}} \left(\frac{\partial F_{\mathbf{R}}(\boldsymbol{\nu}, k)}{\partial \boldsymbol{\nu}} \right) \Big|_{\boldsymbol{\nu} = \hat{\boldsymbol{\nu}}(k-1)}, \quad (29)$$

where $\alpha_{\boldsymbol{\nu}} > 0$ is the step-size parameter. Furthermore, the derivative $\frac{\partial F_{\mathbf{R}}(\boldsymbol{\nu}, k)}{\partial \boldsymbol{\nu}}$ can be expanded into

$$\frac{\partial F_{\mathbf{R}}(\boldsymbol{\nu}, k)}{\partial \boldsymbol{\nu}} = \left(\frac{\partial \boldsymbol{\xi}^T}{\partial \boldsymbol{\nu}} \right) \left(\frac{\partial \mathbf{c}^T}{\partial \boldsymbol{\xi}} \right) \frac{\partial F_{\mathbf{R}}(\boldsymbol{\nu}, k)}{\partial \mathbf{c}}, \quad (30)$$

where the vector $\mathbf{c}$ is defined in (25). The derivatives in (30) consist of three parts, as elaborated next. Due to (22), the derivative $\frac{\partial \boldsymbol{\xi}^T}{\partial \boldsymbol{\nu}}$ is a diagonal matrix. Additionally, the i th diagonal entry of $\frac{\partial \boldsymbol{\xi}^T}{\partial \boldsymbol{\nu}}$ is $\frac{\partial \xi_i}{\partial \nu_i} = (1 - \xi_i^2)/2$. Based on (21) and (3), the derivative $\frac{\partial \mathbf{c}^T}{\partial \boldsymbol{\xi}}$ is a diagonal matrix with the i th diagonal entry

$$\frac{\partial c_{y,i}}{\partial \xi_i} = A \frac{\partial (\xi_i \mathcal{S}(\boldsymbol{\theta}_i(k), \eta_i))}{\partial \xi_i} \quad (31)$$

$$+ \frac{2 \exp(-\bar{\gamma}_k^2)}{\pi} \sum_{n=0}^{L-1} (n+1) e(n) d_k(n) \xi_i^n, \quad (32)$$

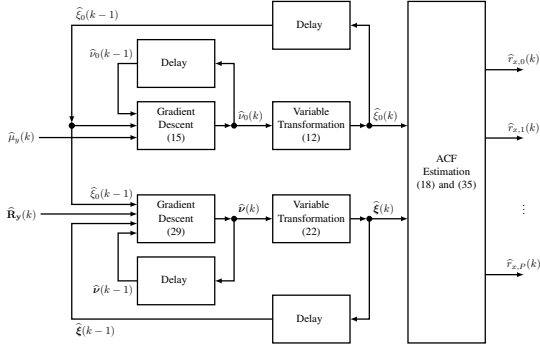


Fig. 2. A block diagram of the one-bit ACF estimators.

where $\eta_i = \beta \xi_i$. The derivative in (31) becomes

$$\frac{\partial (\xi_i \mathcal{S}(\theta_i(k), \eta_i))}{\partial \xi_i} = \mathcal{S}(\theta_i(k), \eta_i) \times \left(\frac{1}{1 - \eta_i^2} + \frac{\eta_i \gamma_{1,k} \gamma_{2,k}}{(1 + \eta_i)^2} + \frac{2\eta_i^2 (\gamma_{1,k} \gamma_{2,k} - \bar{\gamma}_k^2)}{(1 - \eta_i^2)^2} \right). \quad (33)$$

The notations $\gamma_{1,k}$, $\gamma_{2,k}$, $\bar{\gamma}_k$, and $d_k(n)$ are obtained by the expressions of γ_1 , γ_2 , $\bar{\gamma}$, and $d(n)$ in (3) with $\sigma_1 = \sigma_2 = \hat{\xi}_0(k-1)$, respectively. Furthermore, due to (26), we obtain

$$\frac{\partial F_{\mathbf{R}}(\boldsymbol{\nu}, k)}{\partial \mathbf{c}} = 2\mathbf{E}^T (\mathbf{E}\mathbf{c} - \mathbf{b}). \quad (34)$$

Substituting the expressions in (31), (32), (33), and (34) into (30) yields an expression of the gradient.

Next, the original variable $\boldsymbol{\xi}$ in (19) can be updated based on $\hat{\boldsymbol{\nu}}(k)$ and (22). Let $\hat{\boldsymbol{\xi}}(k)$ be the estimate of $\boldsymbol{\xi}$ in the k th block. The i th component of $\hat{\boldsymbol{\xi}}(k)$ is $\hat{\xi}_i(k)$. The signal $\hat{\xi}_i(k)$ and the estimated ACF $\hat{r}_{x,i}(k)$ of $x(n)$ with lag i are

$$\hat{\xi}_i(k) \triangleq \frac{1 - \exp(-\hat{\nu}_i(k))}{1 + \exp(-\hat{\nu}_i(k))}, \quad \hat{r}_{x,i}(k) \triangleq \hat{r}_{x,0}(k) \hat{\xi}_i(k), \quad (35)$$

where $i \in \llbracket 1, P \rrbracket$.

Fig. 2 depicts a block diagram of the proposed one-bit ACF estimators. The adaptive estimators $\hat{\mu}_y(k)$ and $\hat{\mathbf{R}}_y(k)$ are obtained with $\psi(n) = y(n)$ in Fig. 1. The upper branch of Fig. 2 computes the adaptive estimate of the SD $\hat{\xi}_0(k)$. This quantity is delayed and used in the lower branch of Fig. 2 to estimate the normalized ACF $\hat{\xi}(k)$. Based on these quantities, the ACF of $x(n)$ is estimated as $\hat{r}_{x,0}(k), \hat{r}_{x,1}(k), \dots, \hat{r}_{x,P}(k)$.

V. NUMERICAL EXAMPLES

We assume that $x(n)$ is a real-valued autoregressive moving-average (ARMA) Gaussian process specified by

$$x(n) = a_1(n)x(n-1) + g(n) + b_1(n)g(n-1), \quad (36)$$

where $a_1(n)$ and $b_1(n)$ are the ARMA parameters. The white Gaussian noise $g(n)$ has zero-mean and autocorrelation $\mathbb{E}[g(n)g(n-k)] = \sigma_g^2(n)\delta_{k,0}$. The quantization threshold is $\lambda = 1.3$. In the adaptive estimator, we set $P = 3$, $B = 50$, and $N = 50000$. The exponential parameter in (5) is $\tau = 0.99$. The step-size parameters in (15) and (29) are $\alpha_0 = \alpha_\nu = 3$. According to [26], we select $A \approx 0.12475$, $\beta \approx 0.96022$, and $L = 10$. The number of Monte Carlo trials is $M = 1000$.

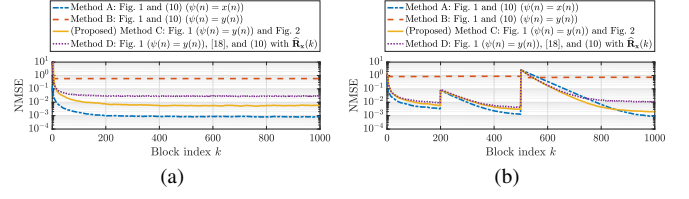


Fig. 3. The NMSE learning curve of adaptive estimators for (a) the stationary case and (b) the non-stationary case.

Let $\hat{r}_{x,i}^{(m)}(k)$ be the estimated ACF with lag i in the k th block and the m th Monte Carlo trial. The corresponding true ACF is $r_{x,i}(k)$. The NMSE in the k th block is defined as

$$\text{NMSE}(k) \triangleq \frac{1}{M} \sum_{m=1}^M \left(\frac{\sum_{i=0}^P |\hat{r}_{x,i}^{(m)}(k) - r_{x,i}(k)|^2}{\sum_{i=0}^P |r_{x,i}(k)|^2} \right). \quad (37)$$

Fig. 3(a) illustrates the NMSE with $a_1(n) = 0.5$, $b_1(n) = 0.3$, and $\sigma_g(n) = 1.5$ for all n . Four methods are based on the combination of Fig. 1, (10), Fig. 2, and the modified arcsine law [18], using $N_m = 10$ samples in the Monte Carlo integration. Method A has the fastest convergence and the lowest error floor, since this estimator is based on the original data $x(n)$. However, Method B exhibits the highest error floor, even for a large k , due to the quantization errors in $y(n)$. The proposed method, Method C, improves Method B through the steps in Fig. 2. Furthermore, Method C exhibits a similar trend to Method A. This attribute results from the calibration of one-bit quantization in Fig. 2. Method C has a lower NMSE floor than Method D, since Method D can lead to undefined power estimates with insufficient data.

Fig. 3(b) depicts the NMSE for a non-stationary $x(n)$. The parameters $(a_1(n), b_1(n), \sigma_g(n))$ are $(-0.9, 0.1, 1.3)$ for $0 \leq n \leq N_1 - 1$, $(-0.7, -0.4, 1.8)$ for $N_1 \leq n \leq N_2 - 1$, and $(-0.4, -0.5, 1.6)$ for $N_2 \leq n \leq N - 1$, where $N_1 = \frac{N}{5}$ and $N_2 = \frac{3N}{10}$. These methods exhibit behaviors close to those in Fig. 3(a). Empirically, Method C estimates the ACF after the change points at $k = 200$ and $k = 500$, with a convergence rate similar to the unquantized ACF estimator in Method A.

VI. CONCLUDING REMARKS

This paper proposed an adaptive ACF estimator based on one-bit quantized Gaussian data. The proposed method is founded on adaptive estimators for the one-bit mean and ACM. These quantities facilitate optimization problems for ACF estimation of the original data. These optimization problems can be solved with gradient descent, making the proposed method suitable for adaptive scenarios.

The proposed method can be extended to non-stationary scenarios where the physical parameters are derived from the ACF. For instance, the proposed method can be generalized to adaptive frequency tracking in time series and adaptive DOA tracking in array signal processing. Future work also includes analyzing the estimation performance of the proposed method.

VII. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical simulation study for which no ethical approval was required.

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